

Web Data Integration Project Abstract (Group 6)

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Abstract. We assess the feasibility of integrating Goodreads, DBpedia, and Amazon book data into a common schema. The sources contain 52,478, 20,981, and 40,000 records, respectively. A blocking key formed from normalized title and first-author surname yields 10,397 distinct keys in at least two datasets and 94,823 distinct keys in the union. A manual review labels 60 of 60 sampled candidate pairs as matches (20 per dataset pair). These are key-based estimates, not a materialized set of confirmed works. ISBN is reserved for gold-standard evaluation and is not used for matching. The target schema contains `title`, `authors`, `genres`, `publication_year`, `publisher`, `page_count`, `language`, and `rating`, `ratingsCount`, `price`, `series`, `awards`; `authors` and `genres` are list-valued.

1 Content

1.1 Use Case

The goal is to integrate book information from three independently maintained sources. They describe overlapping books but use different attribute names, formats, and levels of detail.

The integration aims to identify matching books and combine their information in a common schema with twelve attributes: `title`, `authors`, `genres`, `publication_year`, `publisher`, `page_count`, `language`, and `rating`, `ratingsCount`, `price`, `series`, `awards`.

1.2 Requirements Fulfillment

Datasets Three different datasets are used for this project: a Goodreads dataset from Kaggle, book data from DBpedia, and an Amazon Books dataset from Hugging Face. The Goodreads and Amazon datasets are provided in CSV format. The DBpedia data was retrieved in XML format using a SPARQL query executed in Python. The SPARQL query returned approximately 20,000 results. The DBpedia dataset used in this project contains 20,981 entities.

Table 1 gives an overview of the three datasets, including their sources, formats, number of entities, and available attributes.

The project requirements are addressed as follows:

- **Three different datasets:** Goodreads, DBpedia, and Amazon Books satisfy this requirement.
- **At least 2,500 entities:** The union contains 94,823 distinct normalized blocking keys, comfortably above the threshold. This is a key-based estimate, not a materialized integrated dataset.
- **At least 1,000 entities in at least two datasets:** The key-based estimate is 10,397 distinct keys, above the threshold.
- **At least eight attributes in the integrated dataset:** The planned integrated schema contains 12 attributes and therefore meets this requirement. Books can be identified using a combination of at least two attributes. In particular, `title` and `authors` can be used for this purpose.
- **At least five attributes contained in at least two datasets:** Ten attributes of the planned schema are available in at least two datasets. Seven – `title`, `authors`, `publisher`, `publication_year`, `page_count`, `language`, and `genres` – are represented in all three through the mappings in Table 2. Because DBpedia is sparse for `genres`, `language`, and `year`, fusion by voting across all three sources is mainly planned for `authors`, `publisher`, and `page_count`, which are missing in only 13.5%, 25.4%, and 20.1% of the DBpedia records.
- **List attribute and entity identification:** `authors` and `genres` are list-valued; matching uses normalized `title` and first-author surname. ISBN is excluded from matching and reserved for evaluation.

The datasets also differ in their completeness. Attributes with more than 30% missing values are marked with MV (Missing Values) in Table 1. In the Goodreads dataset, this applies to `series`, `characters`, `edition`, `firstPublishDate`, `awards`, and `setting`. In the DBpedia dataset, `genres` (49.1%), `language` (87.6%), and `year` (88.5%) exceed this threshold. In the Amazon dataset, no attribute exceeds 30%; the highest share of missing values is 23.5% (`isbn10`).

Integrated Schema The source attributes are mapped to canonical names in a common schema (Table 2). Attributes that describe the same information but have different names in the input datasets are mapped to the same attribute. For example, `author` in Goodreads and `authors` in DBpedia and Amazon are mapped to the common attribute `authors`. Goodreads `publishDate`, DBpedia `year`, and the year extracted from Amazon `publication_date` map to `publication_year`; date precision therefore differs by source.

Table 2 shows the planned integrated schema and the corresponding attributes in the input datasets. The planned schema contains 12 attributes. Ten of these attributes are available in at least two datasets. The attributes `title`, `authors`, `publisher`, `publication_year`, `page_count`, `language`, and `genres` can be mapped to attributes from all three datasets.

The shared attributes can later be used to identify matching books across the datasets. For example, a combination of `title` and `authors` can be used for this purpose.

Table 1. Overview of the analyzed datasets.

Dataset	Source	Format	# Ent.	# Attr.	List of attributes
Goodreads Best Books Ever	Kaggle ¹	CSV	52,478	23	title, series (MV), author, rating, description, language, isbn, genres, characters (MV), bookFormat, edition (MV), pages, publisher, publishDate, firstPublishDate (MV), awards (MV), numRatings, ratingsByStars, likedPercent, setting (MV), bbeScore, bbeVotes, price
DBpedia	DBpedia ²	XML	20,981	8	authors, genres (MV), isbns, language (MV), pages, publisher, title, year (MV)
Amazon Reviews 2023	Hugging Face ³	CSV	40,000	13 ⁴	title, subtitle, authors, publisher, publication_date, language, pages, genres, average_rating, ratings_count, price, isbn13, isbn10

¹ <https://www.kaggle.com/datasets/meetnaren/goodreads-best-books>² <https://dbpedia.org/sparql>³ <https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023>⁴ Amazon's 13 attributes exclude the internal identifiers `amazon_id` and `asin`.

Some attributes differ in their representation and level of detail. For example, DBpedia provides only the publication year, while Goodreads and Amazon provide a publication date. Such differences have to be considered during the integration process. **Examples of heterogeneity.** (i) *Titles*: Amazon titles often contain series or edition information in brackets, e.g. “One Small Sacrifice (Shadows of New York, 1)” vs. “One Small Sacrifice” in Goodreads, and capitalization differs (“LETTER TO A CHRISTIAN NATION” vs. “Letter to a Christian Nation”). (ii) *Authors*: Goodreads stores all contributors in one string including roles, e.g. “Terry Brooks (Goodreads Author)”, whereas Amazon stores a serialized list such as ["Jodi Picoult"] and DBpedia uses a list of author elements. (iii) *Language*: DBpedia values read “English language”, while Goodreads and Amazon use “English”.

The datasets also provide different information about books. Goodreads and Amazon contain rating information, while DBpedia does not. Goodreads provides additional information such as series and awards, while Amazon provides a `subtitle` field whose non-empty values (92.9%) mostly contain binding and release information rather than real subtitles. The integration can therefore combine information about a book from different sources.

Table 2. Attribute Intersection with Integrated Schema

Attribute name	Type	Datasets in which the attribute is found
title	string	GR (title), DB (title), AM (title)
authors	list<string>	GR (author), DB (authors), AM (authors)
genres	list<string>	GR (genres), DB (genres), AM (genres)
publication_year	integer	GR (publishDate), DB (year), AM (publication_date)
publisher	string	GR (publisher), DB (publisher), AM (publisher)
page_count	integer	GR (pages), DB (pages), AM (pages)
language	string	GR (language), DB (language), AM (language)
rating	float	GR (rating), AM (average_rating)
ratingsCount	integer	GR (numRatings), AM (ratings_count)
price	float	GR (price), AM (price)
series	string	GR (series)
awards	list<string>	GR (awards)
<i>Identifier for the gold standard only (not used for matching):</i>		
isbn	string	GR (isbn), DB (isbns), AM (isbn13, isbn10)

Notes: One record per *work*. GR = Goodreads Best Books Ever, DB = DBpedia, AM = Amazon; source column names in brackets.

Entity Matching and Overlap Candidate works are blocked by normalized title and first-author surname. Normalization lowercases titles, removes diacritics and punctuation, drops bracketed text and (when present) subtitles, and ignores common leading articles. ISBN does not participate in this key; it

is reserved for the evaluation gold standard. The overlap script reports 6,562 Amazon–Goodreads, 2,129 Amazon–DBpedia, and 4,634 Goodreads–DBpedia shared keys, with 1,464 keys in all three sources. In total, 10,397 keys occur in at least two sources and the union contains 94,823 distinct keys. A manually labelled sample of 20 candidate pairs per source pair contains 60/60 accepted matches (100% sampled precision). These are key-based estimates, not confirmed counts in a materialized integrated database; title normalization can conflate distinct works, and the sample does not measure recall.

Table 3. Overlap summary by distinct normalized blocking key.

Measure	Count
Amazon $\cap$ Goodreads	6,562
Amazon $\cap$ DBpedia	2,129
Goodreads $\cap$ DBpedia	4,634
All three datasets	1,464
Keys in at least two datasets	10,397
Union of distinct normalized keys	94,823

Table 3 counts distinct keys based on normalized title and first-author surname, so it differs from the record counts in Table 1.